Machine Learning Based Analog Circuit Fault Diagnosis

PDF 25 (Supplementary) — Sallen-Key & RC Ladder Circuit Theory

The Electronics Foundation Missing From Most of the Project Bible

SUPPLEMENTARY — Sallen-Key Active Filter & RC Ladder Network

Supplementary Chapter — Sallen-Key Active Filter and RC Ladder Network The Two Circuits Missing From Most of the Project Bible Before Starting Across almost the entire Project Bible, the electronics chapters talked about RC and RLC circuits only. But your actual project uses four circuits. RC RLC Sallen-Key RC Ladder This supplementary chapter fills that gap. It covers exactly what Chapter 5 would have covered if it had included all four circuits from the start.

What you will know after finishing this chapter

After finishing this chapter you should be able to answer [OK] What is a Sallen-Key filter and why is it active? [OK] Why does Sallen-Key need a dual power supply? [OK] What is the cutoff frequency formula and where does it come from? [OK] Why does an RC Ladder network behave differently from a single RC stage? [OK] Where does the mysterious 2.618 constant come from? [OK] How do faults in these two circuits change their waveforms? [OK] Why do these circuits strengthen the generalisation argument of your project?

Difficulty

Basics 4/5

Theory 5/5

Coding 1/5

Viva Importance 5/5

Let's Begin

What is a Sallen-Key filter? Simple definition. The Sallen-Key filter is a second-order active low-pass filter built using an operational amplifier together with two resistors and two capacitors. Question. What does "active" mean here? It means the circuit needs external power to operate. Not just the input signal.

PROFESSOR ASKS

What is a Sallen-Key filter?

EXCELLENT ANSWER

The Sallen-Key filter is a second-order active low-pass filter that uses an operational amplifier with positive feedback through two capacitors and two resistors to achieve a sharper roll-off than a passive RC filter. Why Is This Circuit Different From RC and RLC? Question. Are RC and RLC active or passive? Passive. They only contain resistors, capacitors, and inductors. No power supply. No amplification. Question. What makes Sallen-Key different? It contains an op-amp. That op-amp needs its own power supply. This is the first time in your project that a circuit requires external power beyond the input signal.

PROFESSOR ASKS

Why is including an active circuit important for your project?

EXCELLENT ANSWER

Including an active circuit like Sallen-Key demonstrates that the fault classification pipeline generalises beyond passive networks. Active circuits introduce entirely new failure modes, such as power supply issues and feedback path faults, that passive RC and RLC circuits cannot exhibit.

The Dual Power Supply

Your Sallen-Key implementation uses +15V and −15V supplies for the op-amp (an LT1001 in your simulation). Question. Why two supplies instead of one? Because op-amps typically need both a positive and a negative rail to correctly amplify signals that swing both above and below zero.

Real Life Analogy

Imagine a seesaw. Question. Can it move only upward? No. It needs room to move both up and down around its center. The op-amp's dual supply works the same way.

The Wiring — Where Students Get Confused

This is the single most important detail in the entire Sallen-Key circuit. Question. Where does feedback capacitor C1 connect? Many people assume it connects to ground, exactly like C2. That would be wrong. Correct wiring. C1 connects from the output (Vout) back to the input node (sometimes labelled Node_A). This is called positive feedback, and it is precisely what gives Sallen-Key its sharp, second-order roll-off. Question. What happens if C1 is wired to ground instead? The circuit becomes nothing more than two passive RC stages in series, with an op-amp doing very little useful work. The characteristic Sallen-Key response disappears completely.

PROFESSOR ASKS

Why must C1 connect to the output rather than to ground?

EXCELLENT ANSWER

Connecting C1 from the output back to the input node creates the positive feedback path that defines the Sallen-Key topology. Without this feedback connection, the circuit degrades into two independent passive RC stages and loses its characteristic second-order active response.

The Cutoff Frequency Formula

Your circuit uses

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R1 = R2 = 10 kΩ
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C1 = C2 = 10 nF
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The cutoff frequency is given by fc = 1 / (2 × π × R × C) Substituting your values fc = 1 / (2 × π × 10000 × 10×10^{-9}) fc ≈ 1591.5 Hz Question. How does this compare to a simple RC circuit? A basic RC circuit built from similar-sized components typically has a cutoff frequency closer to 16 Hz. Sallen-Key's cutoff is roughly 100 times higher. Why Is the Rise Time So Much Faster? Question. If Sallen-Key's cutoff frequency is 100 times higher than RC, what happens to its Rise Time? It becomes dramatically faster. Your project's measured Rise Time for Sallen-Key is approximately 0.4 milliseconds, compared to roughly 22 milliseconds for a simple RC circuit.

PROFESSOR ASKS

Why does Sallen-Key settle so much faster than RC?

EXCELLENT ANSWER

Because the Sallen-Key filter has a cutoff frequency roughly one hundred times higher than the RC circuit, its transient response settles proportionally faster, resulting in a much shorter rise time.

How Faults Affect Sallen-Key

Fault 1 — Open Circuit

Suppose R1 fails open, modelled as a very large resistance (around 1 gigaohm). No meaningful signal can pass through the filter. The output stays close to 0V throughout the simulation.

Fault 2 — Short Circuit

Suppose the feedback capacitor C1 fails short, modelled as a very small capacitance (around 1 picofarad). This effectively removes the feedback path that defines the filter's second-order behaviour. The circuit loses its characteristic response and behaves almost like an unfiltered buffer, rising very quickly.

Fault 3 — Component Drift

Suppose R1 drifts from

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10 kΩ
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to

```
15 kΩ.
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Using the cutoff formula, fc drops from about

1592 Hz

to about 1061 Hz. The filter becomes noticeably slower, though it still behaves recognisably like a Sallen-Key filter.

Fault 4 — Noise

Gaussian noise is added on top of the otherwise healthy Sallen-Key response, exactly as with the other three circuits.

PROFESSOR ASKS

How does an open-circuit fault affect the Sallen-Key filter differently from a short-circuit fault?

EXCELLENT ANSWER

An open-circuit fault blocks the signal path entirely, keeping the output near zero volts. A short-circuit fault in the feedback capacitor instead removes the second-order feedback behaviour, causing the circuit to respond much faster and lose its filtering characteristics, rather than blocking the signal. Why the Spectral Centroid Matters More Than Dominant Frequency Here Question.

For Sallen-Key,

which frequency-domain feature tends to be more reliable —

Dominant Frequency

or Spectral Centroid? Spectral Centroid. Because the 20-millisecond simulation window used for Sallen-Key is short relative to its high cutoff frequency, the raw

Dominant Frequency

reading can be imprecise.

Spectral Centroid,

which reflects the weighted average of the whole spectrum, tracks the true cutoff frequency more robustly. PART B — THE RC LADDER NETWORK First Question What is an RC Ladder network? Simple definition. An RC Ladder is two (or more) RC stages connected in series, where each stage feeds into the next. Your implementation uses two identical stages, each with

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R = 1 kΩ
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and C = 10 μF. Question. Is this still a passive circuit? Yes. No op-amp. No external power. Just resistors and capacitors, like the basic RC circuit, but arranged differently.

PROFESSOR ASKS

What is an RC Ladder network?

EXCELLENT ANSWER

An RC Ladder network consists of multiple RC stages connected in series, where each stage's output feeds into the next stage's input, creating a higher-order passive filter through cascaded loading effects. Why Doesn't It Behave Like Two Separate RC Circuits? This is the single most important insight about the RC Ladder. Question. Suppose you have two independent RC circuits, each with τ = RC = 10 ms. If you simply added their responses, what would the combined time constant be? Naively, you might expect something related to 2 × RC. But that is not what happens in a real ladder network. Why? Because

Stage 2

does not just receive

Stage 1's output.

It also loads

Stage 1,

drawing current that changes

Stage 1's

own charging behaviour. The two stages interact. They are not independent.

The Transfer Function

Mathematically, your RC Ladder's transfer function is $H(s) = 1 / ((sRC)^2 + 3(sRC) + 1)$ Question. Why is there a "3" in the middle term, instead of a "2" that you might expect from two independent stages? That extra "1" beyond what independent stages would give you is precisely the mathematical signature of inter-stage loading. It is not a mistake. It is the physics of cascaded networks.

PROFESSOR ASKS

Why does the transfer function contain a coefficient of 3 rather than 2?

EXCELLENT ANSWER

The coefficient of 3 arises because the second RC stage loads the first stage, drawing current that alters its charging behaviour. This inter-stage loading effect distinguishes a true cascaded RC ladder from two independent, buffered RC stages, which would instead produce a coefficient of 2. Where Does 2.618 Come From? This is one of the most elegant results in your entire project. Solving the quadratic $(sRC)^2 + 3(sRC) + 1 = 0$ for its roots gives two poles. $sRC = (-3 \pm \sqrt{5}) / 2$ which evaluates to approximately −0.382 and −2.618 Question. Which pole dominates the response? The slower pole, meaning the one closer to zero — in this case, −0.382. The effective time constant of the whole ladder becomes $\tau_{eff} = RC / 0.382 \approx 2.618 \times RC$ Question. With

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R = 1 kΩ
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and C = 10 μF, what is τ_{eff} numerically? $\tau_{eff} \approx 2.618 \times 10$ ms ≈ 26.2 ms

PROFESSOR ASKS

Where does the value 2.618 come from in your RC Ladder analysis?

EXCELLENT ANSWER

The value 2.618 emerges from solving the characteristic equation of the two-stage RC Ladder's transfer function. It is the reciprocal of the dominant pole magnitude, 0.382, which governs the slowest decaying mode of the network and therefore its effective time constant.

Why Rise Time Is Roughly 58 Milliseconds

Question. Using the standard approximation Rise Time ≈ 2.2 × τ, what

Rise Time

would you expect for the RC Ladder? 2.2 × 26.2 ms ≈ 57.6 ms Your project's measured value is approximately 58.7 ms, which agrees closely with this theoretical prediction. Why the Waveform Has an S-Shape Question. Does the RC Ladder's step response look like a simple RC exponential curve, or something different? Something noticeably different — an S-shaped curve. Question. Why? Think about what

Stage 2

experiences at the very start, at t = 0.

Stage 1's

capacitor has not yet charged at all. So

Stage 2

initially receives almost no driving voltage. Its own output starts completely flat, then gradually begins to rise only once

Stage 1

has charged enough to start driving it. This delayed, gradual acceleration followed by a more familiar exponential-style settling is exactly what produces the visible S-shape. A single-stage RC circuit, by contrast, starts rising immediately and as fast as it ever will, right from t = 0.

Real Life Analogy

Imagine pushing a line of people, one behind the other, to start walking. Question. Does the person at the very back start moving at the exact same instant as the person at the front? No. The back person's motion is delayed and gradual, depending on everyone ahead of them first getting moving. That delayed, gradually-accelerating start is the S-shape. How Faults Affect the RC Ladder

Fault 1 — Open Circuit

If R1 (the first stage's resistor) fails open, no meaningful current can flow into the ladder at all. Both stages remain close to 0V. Fault 2 — Short Circuit — The Most Interesting Case Suppose C1 (Stage 1's capacitor) fails short, collapsing to around 1 picofarad. Question. What happens to

Stage 1?

Its time constant collapses to almost instantaneous, effectively vanishing as a meaningful delay element. Question. But is

Stage 2

also destroyed? No.

Stage 2

remains completely healthy, with its own normal

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R = 1 kΩ,
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C = 10 µF. Question. So what does the overall output look like? Approximately a single-stage RC response, with

Rise Time

close to 22 ms — not the instantaneous jump you might expect from a short, and not the slow, 58 ms, S-shaped response of a healthy two-stage ladder either. This is called order reduction — the faulty two-stage system behaves like a lower-order, single-stage system.

PROFESSOR ASKS

Why does a short circuit in the first stage of the RC Ladder not cause an instantaneous output?

EXCELLENT ANSWER

Although the short circuit effectively removes the first stage's time constant, the second stage remains fully functional with its own resistor and capacitor. The overall response therefore reduces to that of a single healthy RC stage, producing an intermediate rise time rather than an instantaneous jump or the original slow two-stage response.

Fault 3 — Component Drift

If R1 drifts from

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1 kΩ
```

to

```
1.5 kΩ,
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both the effective time constant and the loading interaction with

Stage 2

are affected, making the overall response slower than the healthy case on both fronts simultaneously.

Fault 4 — Noise

As with every other circuit, Gaussian noise is added on top of the otherwise healthy S-shaped response. Why This Matters for the Confusion Matrix Remember from

Chapter 28

that RC_Open was confused with RLC_Open, because both settle near 0V and share similar steady-state behaviour. Question. Does the RC Ladder introduce a similar risk? Yes. RC_LADDER_short (order-reduced to roughly 22 ms

Rise Time)

can resemble plain RC_short superficially, since both eventually involve a fast-charging single-stage-like behaviour. This is exactly the kind of cross-circuit confusion that motivated DC Level and the

Topology Rule

in the first place — except here it would be cross-circuit confusion between RC and RC_LADDER, rather than between RC and RLC. Why These Two Circuits Strengthen Your Project's Generalisation Argument Recall the generalisation concern your supervisor raised — that the project needed circuits general enough to prove the methodology works broadly, not just on one specific circuit. Question. How do Sallen-Key and RC Ladder address this concern directly? Sallen-Key introduces an active, op-amp-based topology, with entirely new failure modes (power supply issues, feedback path breaks) that no passive circuit can exhibit. RC Ladder introduces a multi-stage, higher-order passive topology, with inter-stage loading effects and order-reduction behaviour that a single-stage RC circuit cannot exhibit. Together with RC (first-order passive) and RLC (second-order passive, resonant), your four circuits span passive vs active, single-stage vs multi-stage, and non-resonant vs resonant behaviour — a genuinely broad test of whether your 17-feature, SVM-based methodology generalises, rather than simply overfitting to one specific circuit's quirks.

PROFESSOR ASKS

Why did you include four different circuit topologies instead of just one?

EXCELLENT ANSWER

Including RC, RLC, Sallen-Key, and RC Ladder circuits allowed the methodology to be tested across passive and active topologies, single-stage and multi-stage networks, and resonant and non-resonant behaviour. This demonstrates that the feature extraction and classification pipeline generalises across fundamentally different circuit architectures, rather than being tuned to the characteristics of a single circuit.

COMMON MISTAKES

MISTAKE: [X] Thinking Sallen-Key's feedback capacitor connects to ground. Wrong. It connects from the output back to the input node, creating positive feedback.

MISTAKE: [X] Thinking the RC Ladder's time constant is simply 2 × RC. Wrong. Inter-stage loading produces an effective time constant of 2.618 × RC, not a simple doubling.

MISTAKE: [X] Thinking a short circuit in RC Ladder's first stage produces an instantaneous output. Wrong. The second stage remains healthy, producing an order-reduced, single-stage-like response instead.

MISTAKE: [X] Thinking Sallen-Key is passive because it looks similar to RC circuits on a schematic. Wrong. The op-amp and its dual power supply make it fundamentally active.

REVISION NOTES

Remember these key facts. **Sallen-Key** — Active,. second-order, fc = 1/(2πRC) ≈ 1592 Hz, needs dual power supply, C1 feeds back to output, not ground. **RC Ladder** — Passive,. multi-stage, τ_eff = 2.618 × RC ≈ 26.2 ms, inter-stage loading produces the "3" coefficient, S-shaped step response, short-circuit fault causes order reduction, not instantaneous response.

MEMORY TRICK

For Sallen-Key, remember "Feedback to output, not ground — that's what makes it Sallen-Key." For RC Ladder, remember "2.618, not 2 — because the stages load each other."

KNOWLEDGE CHECK

Level 1

- What is a Sallen-Key filter?
- Why does it need a dual power supply?
- What is an RC Ladder network?
- Why is the RC Ladder's time constant not simply 2×RC?
- What causes the S-shape in the RC Ladder's response?

Level 2

- Why must C1 connect to the output rather than to ground in Sallen-Key?
- How does an open-circuit fault differ from a short-circuit fault in Sallen-Key?
- Where does the value 2.618 come from mathematically?
- Why does a short circuit in RC Ladder's first stage not produce an instantaneous output?
- Why is Spectral Centroid more reliable than Dominant Frequency for Sallen-Key?

Professor Level

- Explain why including Sallen-Key and RC Ladder strengthens the generalisation argument of your project.
- Derive the effective time constant of the RC Ladder from its transfer function.
- Compare the fault behaviour of RC_Open and RC_LADDER_short and explain why they might still be confused by the classifier.
- Why does Sallen-Key's cutoff frequency being 100 times higher than RC's matter for feature extraction?
- If you added a third RC Ladder stage, how would you expect the effective time constant coefficient to change?
- End of Supplementary Chapter

Congratulations. You now have real circuit theory for all four topologies used in your project, not just RC and RLC. Combined with the rest of the Project Bible, you can now explain the complete electronics foundation of your work — including the two circuits that most of the Bible accidentally left out.